FinTech Innovation Challenge

Pulse

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Problem Statement & Hypothesis

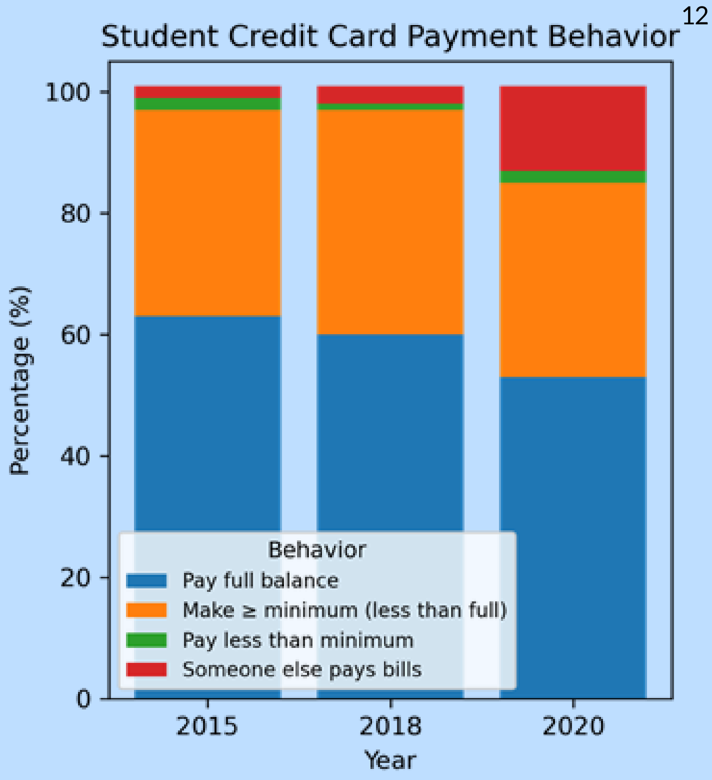
- Students don't fully understand credit cards or their *long-term consequences*[7].
- Digital payments make spending feel *"invisible"*, making it easier to overspend[13].
- Financial apps largely fail young adults who are still learning:
    - Budgeting apps focus on past spending, but *don't shape decisions or curb bad habits.*
    - Credit card platforms only look at potential benefits and *assume ideal user behavior*

Providing *real-time*, *actionable interventions* and card recommendations based on *how students actually manage debt* will help students **spend smarter and avoid costly mistakes.**

Research

- 42% of students **carried credit card debt** in 2023[7]
- 46% went into debt from **non-essential spending**[7]
- Students **spend 2× more** with credit cards than cash[13]
- **Only 11% understand** interest rates and credit[14]

Credit cards *drive overspending*, and lacking financial literacy, students easily slip into **costly credit card debt**.



Persona

Emma

“The Dependent Discoverer”

*Image generated by Google Gemini on 11/19/25

Bio

18-year-old freshman experiencing financial independence for the first time She has an authorized user credit card on her parent’s account for “emergencies,” but uses it for daily wants. Despite being on the full meal plan, she frequently orders delivery because digital payments feel abstract since she has never earned money before. She avoids checking her balance, and was shocked when her first statement was $1340 for what felt like small purchases to her.

Goals

- Maintain good grades and make close friends during her first year
- Prove to her parents she can handle the responsibility of having a credit card

Motivations

- Desires to feel “adult” and independent for the first time
- Values her parents’ trust and doesn’t want to disappoint them

Frustrations

- Feels guilty after spending but doesn’t know how to stop the cycle
- Has no reference point for which type of purchase is reasonable

Why she needs our service

Emma needs **real-time awareness** at the moment of purchase because she has no financial literacy and no income to anchor her spending. Our solution provides immediate feedback in terms translatable to what she can understand: “You just spent $25 ordering food on Ubereats! That’s 3 meal swipes you already paid for!” Location-based alerts catch her before impulse purchases so she is able to change her behavior before it is too late.

Persona

Jordan
"The Overconfident Earner"

*Image generated by Google Gemini on 11/19/25

Bio

Jordan is a 21-year-old junior who world as a campus tour guide earning around $360/month. He lives off-campus and pays majority of his own rent, which made him feel ready for his first credit card. However, he confuses gross income with disposable income, and doesn't realize that after his fixed expenses, he is in the negatives before any discretionary spending. He uses Apple Pay exclusively, where spending feels validates because he feels he "earned it" by working.

Goals

- Graduate with a marketing degree and prove financial independence
- Build his credit score and establish himself as a responsible adult

Motivations

- Pride in earning his own money and not relying on parents
- Wants to enjoy college while he's working hard to pay his way through

Frustrations

- Confused why he is broke despite having a job
- His credit score is low because he is unable to make payments on time

Why he needs our service

Jordan represents the most dangerous user type, as he thinks he's financially responsible, but fundamentally misunderstands cash flow. Our solution provides immediate feedback: "You just spent $85 at Lululemon. That is 6.5 hours of work being a tour guide." This notification helps him understand that his paycheck isn't "extra money." He desperately needs intervention before the statement shock because he doesn't even realize he needs the help until it's too late.

Design Thinking

Empathy

Understanding students' real pain points

• Conducted interviews and observation to understand why students overspend

• Recognized that students need support *in the moment*, not after the fact.

• Mapped moments of highest risk (mall, food delivery, social spending) to inform targeted interventions.

Ideation

Generating solution to meet students' needs

• Brainstormed real-time, behavior-shaping features.

• Explored ways to simplify complex credit concepts.

• Prioritized features that create immediate awareness rather than adding budgeting "homework."

Prototype

Testing and bringing solutions to life

• Built early versions of proactive alerts, spending summaries, and translations.

• Created low-fidelity mockups to validate tone, timing, and clarity of interventions.

• Iterated based on student feedback, ensuring alerts felt supportive, not judgmental.

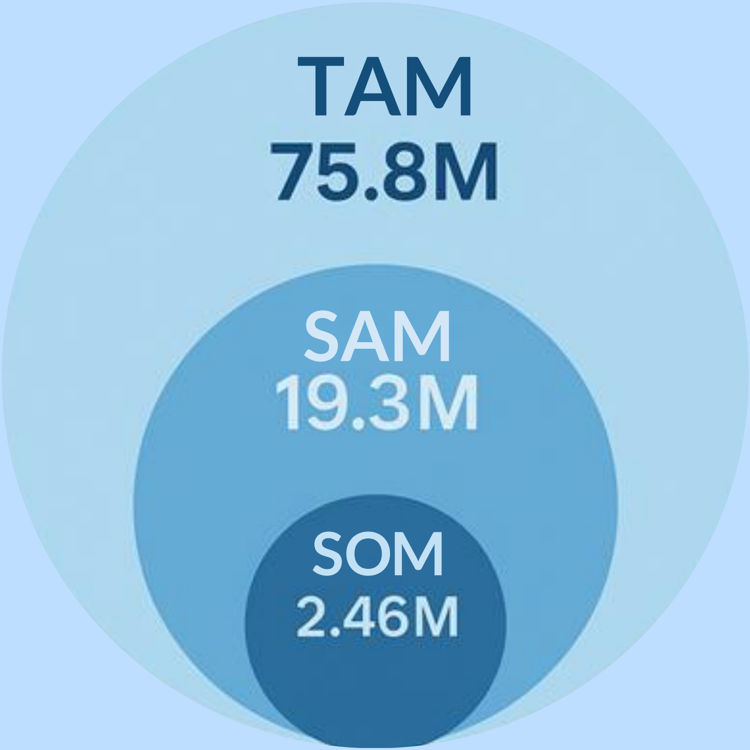
Landscape Analysis

	Monarch[1, 2]	Quicken Simplifi[1, 2]	Origin[1, 2]	Rocket Money[1, 2]
Product Features	• Set financial goals • Track spendings • See your net worth • Manage investments	• Track spendings • View upcoming bills • Groups transactions • Simple budget features	• All-in-one money management app • AI financial coaching • Manage investments • Create will/trust[1]	• Calendar budget • Track spendings • Net worth tracking • Set spending goals
Strengths	• Clean UI/UX • Connect all accounts • Separate login for family members[1]	• Good visuals • Easy to use mobile app • Monitors credit score	• Includes tax software • Best for B2B • Multi-currency	• Free version available • Monitors credit score
Weaknesses	• Built for adults with complex finances	• Assumes steady income • Slow to update • Limited debt management tools	• Limited goal tracking	• Not great for irregular income (especially for students)

Key Takeaways

1. **No real-time, behavior-shaping spending guidance**
2. Existing apps are **built for adults with steady income and complex finances**
3. Most tools focus on **tracking past spending**, not preventing bad habits or overspending before it happens.
4. There is a **large, underserved student market**

Market Size



TAM[10, 16]

All U.S. young adults (18–35)

18–19 (8.7M), 20–24 (21.5M), 25–29 (22.4M), 30–34 (23.1M)

SAM[6, 7, 9, 12]

All undergraduate students in U.S. colleges

~52.7% have credit cards

SOM[15]

All freshmen in U.S. colleges

~1.8M from 4-year programs

~0.66M from 2-year programs

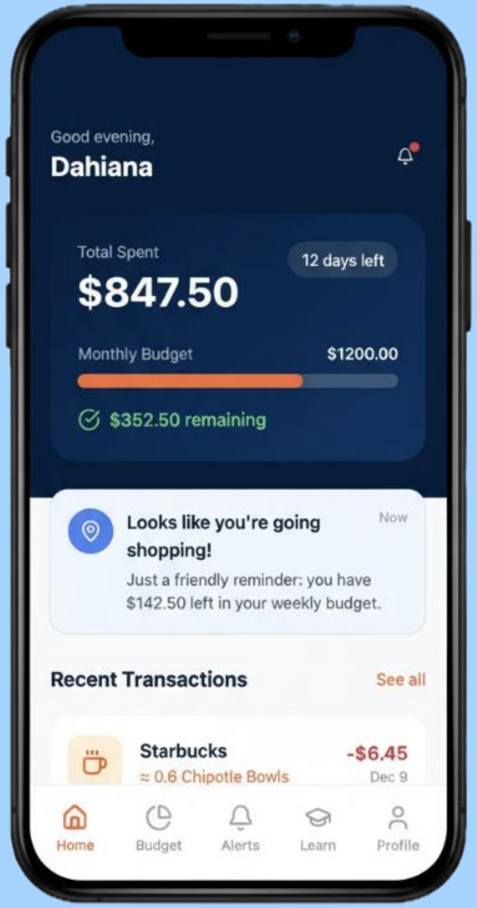
Ages 18–19

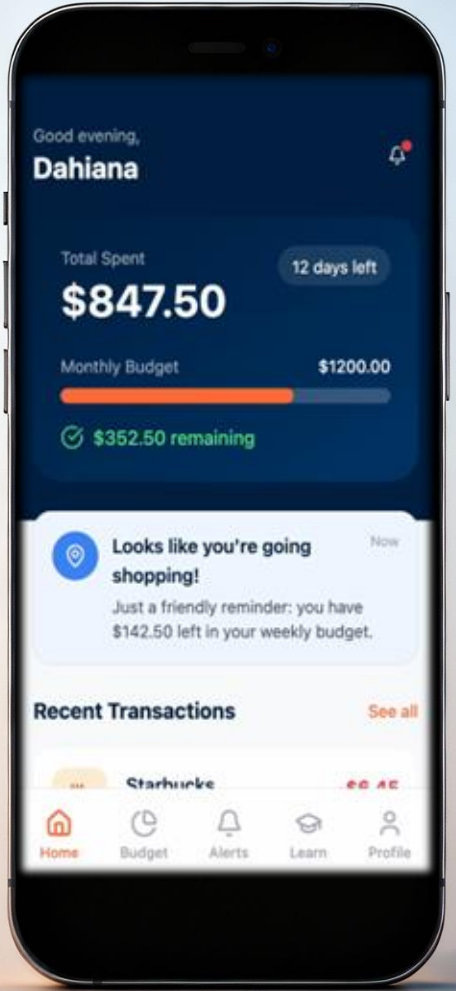
growing +5.5% YoY

Solution

Pulse helps students make smarter financial decisions by giving them **real-time spending guidance** and **behavior-based card recommendations.**

- ✓ Proactive spending alerts
- ✓ Spending summaries
- ✓ Meaningful, student-friendly translations
- ✓ Card recommendations
- ✓ Relevant, AI-driven learning




Good evening,
Dahiana
Total Spent
$847.50
12 days left
Monthly Budget
$1200.00
$352.50 remaining
Looks like you're going shopping!
Now
Just a friendly reminder: you have $142.50 left in your weekly budget.
Recent Transactions
See all
Home
Budget
Alerts
Learn
Profile

Business Case for Management

A habit-building, student-first solution gives a competitive edge in the first-card market.

Guided, lower-risk usage leads to more predictable spend and interchange fees of up to 1-3% per transaction[11].

Capturing students early drives long-term value: a 5% retention lift can raise profits by up to 95%[5].

Implementation Plan

Months 0-6

- Build features
- Pilot project with Tufts

Months 7-18

- Scale to all universities in Boston area
- Refine features
- Add parent dashboard

Months 19-24

- National rollout
- Continue to optimize AI model

Approach

"Student Finance Coach" application

- Real-time alerts
- Location tracking
- Spending translations
- Budget tools
- Credit education

Integrations

- Transaction data
- Location services API
- AI/ML personalization
- Push notifications

Risks and Mitigation

	Impact	Mitigation	Supporting Data
Market resistance due to tightening credit standards	22% decrease in student cardholders [7] Harder to acquire student customers if banks are cautious about lending	Our solution makes banks more confident to issue cards to students by providing structure Reduced competition = easier market entry	52.7% of students still have credit cards [7] → market is large despite decline
Lack of trust in AI	49% concerned about transparency in AI decision-making [15]	83% trust banks → leverage trusted reputation [15] AI featured can be disabled	In personal finance, 51% of people see AI's potential in enhancing financial decisions [15]
Notification Fatigue	Gen Z users receive hundreds of notifications per day Without engagement, the intervention doesn't happen at moment of temptation	Smart AI notification system: Our AI model learns from user behavior patterns based on how they respond to various timing, wording, and response rates Optimizes alert delivery for max effectiveness for each user	60% of users engage more frequently with notified apps Personalized notifications resonate more with users by making them feel understood, which increases interaction and fosters loyalty to platform
Difficult to break impulse spending habits	46% of students got into debt through impulsive and non-essential purchases [7]	Behavioral approach at moment of temptation: · Location alerts when entering stores · Spending translations make abstract purchases concrete · Real-time budget feedback · Pattern insights help identify spending triggers	Discover published an article explaining the main steps to decreasing impulse spending are get to the root cause, budget for discretionary spending, reconsider payment methods, set spending rules [4] → our solution implements them all

Next Steps to Execute

Funding Approval

Mobile App Development

AI/ML model training and integration

Beta user incentives and recruitment

Legal/Compliance

Terms of service for AI-powered financial guidance

Privacy policy review for location-based services

Strategic Partnerships

Partnership approval with Tufts University

Integration pathway into our bank's mobile banking platform

Marketing Resources

Co-marketing budget for student outreach ($5K)

Social media campaign support for beta recruitment

Thank You!

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